# Aayush Kumar Chaudhary

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**Research Interests**: Image Processing, Computer Vision, Eye tracking, Machine Learning (Deep Learning), Human Computer Interaction

## EDUCATION

**Ph.D. student,** Chester F. Carlson Center for Imaging Science — **2016 – Present**
Rochester Institute of Technology — Rochester, New York
**Area of focus**: eye tracking methodology (image processing and machine learning)
Relevant courses: image processing and computer vision, deep learning for vision, AI explorations, Multiview geometry, human visual system, Bayesian statistics for machine learning

**Bachelor of Engineering (B.E.),** Electronics and Communication Engineering, Pulchowk Campus, I.O.E., Tribhuvan University — **2010 – 2014**
Kathmandu, Nepal
**(University Rank: 1st 85.47%)** [Engineering Stream]
**Thesis:** Teleoperation of Mobile Robotic Arm Through Imitation of Human Arms Movement

## SKILLS

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|---|---|
| **Programming languages** | **Python,** Matlab, C, C++ |
| **Packages** | **PyTorch, TensorFlow, Scikit-Learn, OpenCV,** VisualSFM |
| **Eye trackers** | Pupil Labs eye trackers, custom-made remote eye trackers with mirrorless cameras |
| **Miscellaneous** | Human subjects research, Robotics, Circuit design tools (Kicad, Proteus), photography (iris) |

## RESEARCH EXPERIENCE

**Multidisciplinary Vision Research Laboratory, RIT, Research Assistant**
**Advisor: Dr. Jeff Pelz.** — **June 2017 – Present**

**Gaze Estimation**
- Study of feasibility of iris features based gaze estimation.
- Developed a new approach of incorporating regression-based models with machine learning based approaches.
- Improved the precision and accuracy of eye trackers.
- Designed and tested eye tracking hardware setup for study of various eye movements for human subjects.

**Segmentation of eye-regions**
- Improved the robustness of eye-parts segmentation with memory efficient models.
- Designed novel framework for ellipse segmentation on eye-parts
- Currently exploring temporal segmentation

**Privacy in eye videos**
- Designed concept for a novel approach to preserve privacy in the eye videos without degrading the accuracy of eye trackers.

**Synthetic dataset for eye-tracking applications**
- Collaborated with multiple research labs in RIT to introduce a synthetic eye image generation platform for eye-tracking applications. (https://cs.rit.edu/~cgaplab/RIT-Eyes/)

## PUBLICATIONS

JOURNAL ARTICLES

[J.2]* *[Currently in review]* **Chaudhary, A. K.** & Pelz, J.
$\pi_t$:Enhancing the Precision of Eye Tracking using Iris Feature Motion Vectors

[J.1] ***[JEMR 2019]*** *Journal of Eye Movement Research* **Chaudhary, A. K.** & Pelz, J.
Motion tracking of iris features to detect small eye movements.

CONFERENCE PROCEEDINGS

[C.6] *[arXiV] (* equal contribution)* Kothari, R.*, **Chaudhary, A. K.*,** Bailey, R. J., Pelz, J., Diaz, G.
EllSeg: An Ellipse Segmentation Framework for Robust Gaze Tracking

[C.5] ***[ACM-SAP 2020]*** *(to appear)* *ACM Symposium of Applied Perception* Nair, N., Kothari, R., **Chaudhary, A. K.,..,** Bailey, R. J.
RIT-Eyes: Rendering of near-eye images for eye-tracking applications

[C.4] ***[ACM-ETRA 2020]*** *(short-paper)* *Eye Tracking Research & Applications* **Chaudhary, A. K.** & Jeff B. Pelz. 2020.
Privacy-Preserving Eye Videos using Rubber Sheet Model applications

[C.3] ***[ACM-ETRA 2020]*** *(adjunct)* *Eye Tracking Research & Applications* Nair, N., **Chaudhary, A. K.,** Kothari, R., Diaz, G., Pelz, J., Bailey, R.
RIT-Eyes, realistically rendered eye images for eyetracking applications

[C.2] ***[ACM-ETRA 2019]*** *Eye Tracking Research & Applications* **Chaudhary, A. K.**
Motion tracking of iris features for eye tracking.

OPEN CHALLENGES

***[ICCVW-2019]*** *IEEE/CVF International Conference on Computer Vision Workshop* **Chaudhary, A. K.*,** Kothari, R.*, Acharya., M.*, Dangi., S,.., Pelz., J.
RITnet: Real-time Semantic Segmentation of the Eye for Gaze Tracking
***[Semantic Segmentation Challenge: Winner]*** *(oral)*

## WORK EXPERIENCE

**Chester F. Carlson Center for Imaging Science, RIT** **Rochester, New York**
Teaching Assistant (Undergraduate Courses Assisted: Vision and Psychophysics, Linear Methods of Imaging) Aug 2016-May 2017

**Samsung India Electronics Pvt. Ltd (SIEL, Nepal branch)** **Kathmandu, Nepal**
Senior Engineer (Smart phones performance testing and validation (METI team)) Nov 2014-July 2016

- My contribution in Q1'2016 was recognized with a performance award.

## SCHOLARSHIPS, AND AWARDS

*Professional Achievement*

2016 **Performance Award in recognition of High Performance for Q1'2016**, Samsung (SIEL)

*Academic Achievement*

2019 **Prof. F.N.Trofimenkoff academic achievement award,** a prestigious award given to one student each year by Department of Electronics and Computer Engineering, Pulchowk Campus, IOE (DoECE) for academic success during undergraduate studies.

2019 **Winner of Facebook OpenEDS competition.** Our model is memory efficient and achieves the state-of-the-art results on the 2019 OpenEDS Semantic Segmentation challenge.

2015 **NCell Excellence Award,** an award for being the overall topper of DoECE

2010/14 **University Rank 1st (Engineering Stream),** Rank First in All Bachelor's Degree programs of I.O.E. Tribhuvan University.

2014 **NCell Scholarship Award,** Third-year topper of DoECE

2010 **Nepal Board Topper,** Scored the highest score in the country in Higher Secondary Education Board (12th grade) Science faculty.

*Merit Scholarship*

2016- **RIT Ph.D. Scholarship/Assistantship,** Financial Support for Ph.D. studies at RIT.

2010/14 **College Fellowship,** Awarded for outstanding academic performance in each semester of Bachelor of Engineering by I.O.E., Pulchowk Campus.

2010/14 **Full Academic Scholarship, a**warded to pursue B.E. at I.O.E, Pulchowk Campus based on nationwide competitive examination.

## Activities and Leadership

2014 **Coordinator, LOCUS 2014 (Technological Event)**

- Coordinated and organized the 11th national technological festival- the biggest technological festival of the country for DoECE and Department of Electrical Engineering.

2013/14 **Secretary, Amateur Radio Club,** A licensed Ham-Radio Operator

2013 **Team Member, ABU Robocon 2013, Vietnam**

- Integral part of the hardware team (designing and building circuit boards) for the both manual and automatic robot that represented Nepal in the Asia-Pacific Broadcasting Union (ABU) Robocon 2013, Vietnam.

2013 **Secretary, Electronics Club**

- Organized crash course to freshmen regarding Electronics circuits and helped to launch the first magazine of Electronics Club 'Graphene'.

2018 **IdeaLab -** Presented TUBBLE, an engaging Toy for Active Kids.

2014 **Child App Competition**

- Presented a social interacting app 'Beautiful Minds' for autistic children and was one of the finalists of the Child App Competition. It was organized by Microsoft Innovation Center (Nepal) and UNICEF Nepal.

## Miscellaneous

**[Poster.3]** **RITnet: Real-time Semantic Segmentation of the Eye for Gaze tracking**

- ICCV 2019, Workshop on Eye Tracking for VR and AR

**[Poster.2]** **Motion Tracking of Iris Features for Eye Tracking.**

- In 2019 Symposium on Eye Tracking Research and Applications (ETRA'19)

**[Poster.1]** **Motion Tracking of Iris Features to Detect Small Eye Movements.**

- Graduate Showcase, RIT, 2018

**[Talk.1]** **RITnet: Real-time Semantic Segmentation of the Eye for Gaze tracking.**

- Invited talk, in ICCV 2019, Workshop on Eye Tracking for VR and AR.

**[Reviwer.2]** **Eye Tracking Research and Applications (ETRA).**

**[Reviwer.1]** **Journal of Eye Movement Research (JEMR).**

**[Judge.1]** **Judge, LOCUS 2016 (Technological Event)**

- Represented SIEL, Nepal branch as one of the main judges of the event Samsung Galaxy App Quest conducted to promote application development in Nepal.

**[Mentor]**

- High school students (219, 2020)